Lecture 25 — Simulation and long-run behavior

Wed Oct 21

Last updated: October 8, 2026.

Boyd video (optional): Lecture 26 — linear dynamical systems · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Simulation of linear dynamics (class notes / Boyd 9.1–9.3 continued). Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Note: Today we **observe** long-run patterns by simulating. We do **not** yet explain them with eigenvalues — that mystery returns in Week 13.

Learning objectives

By the end of class, students should be able to:

1. Simulate many time steps computationally.
2. Plot state components through time.
3. Recognize stabilization, growth, decay, or oscillation.
4. Conduct a simple what-if simulation.
5. State limitations of a linear dynamical model.

Warm-up

With

$$A = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

compute x_1 and x_2 by hand. Guess (no calculator marathon): after many steps, will x_t blow up, shrink to zero, or settle somewhere nonzero?

Fill in

Simulation recipe (any language):

1. Choose A and initial state x_0.
2. For $t = 0, 1, \ldots, T-1$: set $x_{t+1} =$ _____.
3. Store or plot the components of $x_0, x_1, \ldots, x_T$ versus time.

Long-run patterns (names only — recognize from plots / tables):

Pattern	What you see
Stabilization	x_t approaches a fixed vector (or settles in magnitude)
Growth	components grow without bound (often roughly geometric)
Decay	components shrink toward _____
Oscillation	components bounce / cycle in sign or value

What-if. Change one thing — a parameter inside A, or the initial state x_0 — and re-simulate. Compare the two trajectories.

Limitations of $x_{t+1} = Ax_t$. The model is _____ (no x^2 terms, no products of states); A is assumed constant; time is discrete; there is no noise, capacity limit, or policy change unless you build it into A. Real systems may only match the model for a while.

Worked in class

W1. Many steps on the board / laptop. Let

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \qquad x_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Compute (or run) x_1 through x_5. Sketch each component versus t. Does the trajectory look like growth, decay, oscillation, or stabilization?

W2. Recognize the pattern. Without naming any fancy theory, match each matrix to the most plausible long-run story for a generic nonzero x_0:

(a) $A = \text{diag}(1.2,\ 0.5)$

(b) $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

(c) $A = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$

W3. What-if. Same A as W1, but start from $x_0 = (0, 1)$ instead of $(1, 0)$. Compute x_1 and x_2. Do the two initial conditions seem headed toward the *same* long-run mix, or different ones?

W4. Limitations. A linear infection model $x_{t+1} = Ax_t$ with fixed A predicts infections forever according to the same rule. Name two real-world reasons this could fail after a few weeks (policy, behavior, medicine, saturation, …).

Your turn

Y1. Simulation plan. Write 3–4 lines of pseudocode (or Octave/Python if you prefer) that, given A and x_0, prints $x_0, x_1, \ldots, x_{10}$.

Y2. Pattern ID. You simulate $x_{t+1} = Ax_t$ and obtain the first component sequence

$$3,\ -3,\ 3,\ -3,\ 3,\ \ldots$$

Which pattern name fits best? Give one 2×2 matrix that could produce oscillation like this from $x_0 = (3, 0)$ (many answers OK; one simple choice is enough).

Y3. What-if. Let $A = \text{diag}(0.5,\ 0.5)$ and $x_0 = (8, 2)$.

(a) Compute x_1 and x_2 by hand.

(b) If instead $x_0 = (80, 20)$, how do x_1, x_2 compare? What limitation / feature of *linearity* does this illustrate?

Y4. Conjecture (no proof needed). For

$$A = \begin{bmatrix} 1.1 & 0 \\ 0 & 0.5 \end{bmatrix},$$

compare long-run behavior starting from $x_0 = (1, 0)$ versus $x_0 = (0, 1)$. Write one sentence you want explained later in the course.

Answers (Your turn)

Y1. Example pseudocode:

```
x ← x0
print x
for t = 1 to 10:
    x ← A x
    print x
```

(Octave: `x = A*x` in a `for` loop; Python: `x = A @ x`.)

Y2. **Oscillation.** One simple choice: $A = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$ (or $A = -I$, or a 180° rotation), which sends $(3, 0) \mapsto (-3, 0) \mapsto (3, 0) \mapsto \cdots$.

Y3. (a) $x_1 = (4, 1)$, $x_2 = (2, 0.5)$. (b) $x_1 = (40, 10)$, $x_2 = (20, 5)$ — exactly $10\times$ the previous trajectory. Linearity scales the whole path with the initial state; the model has no "saturation" that would break that scaling.

Y4. Sample conjecture: from $(1, 0)$ the state grows without bound along the first axis; from $(0, 1)$ it decays to zero — so long-run fate seems to depend on the *direction* of x_0, not just its size. (Week 13 will explain this with eigenvectors.)